BC 3 Name: _

Taylor Series Review

1. Use Maclaurin series to write power series expanded about $x_0 = 0$ (in simplified Σ-notation) for each of the following functions.

   a. $f(x) = \dfrac{3x}{2-5x}$

   b. $f(x) = \dfrac{x^3}{e^{2x-1}}$ (Be careful with this one!)

2. Write the first four nonzero terms of the Taylor series for $f(x) = \sin(x)$, expanded about $x_0 = \dfrac{3\pi}{4}$.

3. Find an upper bound for the error made in approximating the value of $e^{\sqrt{3}}$ using the nonzero terms through $n = 6$ in its Maclaurin series.

4. When approximating $\ln(0.8)$, how many nonzero terms of its Maclaurin series do you have to use in order to have an error that is less than 0.0001?

5. What are the possible values for x if we want the error in approximating $\cos(2x)$ using the first four nonzero terms of its Maclaurin series to be less than 0.01?

6. When approximating $\tan^{-1}(0.75)$, how many nonzero terms of its Maclaurin series do you have to use in order to have an error that is less than 0.0001?

7. Suppose $f(2)=1,\ f'(2)=2,\ f''(2)=4,\ f'''(2)=18,$ and $\left|f^{(n)}(x)\right|<6$ for all x and for $n > 3$.

   a. Write the first four nonzero terms of the Taylor series for f expanded about $x_0 = 2$.

   b. Find an upper bound for the error made in approximating the value of $f(2.3)$ using the first four nonzero terms of Taylor series for f expanded about $x_0 = 2$ from part (a).

8. Use Maclaurin series to evaluate $\lim_{x\to 0} \dfrac{\ln(1+x)}{\sin(3x)}$.

9. Use power series to approximate the value of $\int_0^{0.9} \sin\left(x^2\right) dx$ with error less than 0.0001.

10. Suppose that f is a function such that $f(1) = 1,\ f'(1) = 2,$ and $f''(x) = \dfrac{1}{\left(1 + x^3\right)}$ for all $x > -1$.

    a. Estimate the value of $f(1.5)$ using a quadratic Taylor polynomial.

    b. Determine an upper bound for the error made in your approximating in part (a).

11. We can show that the series $\sum_{n=0}^{\infty} \frac{n^2}{2^n}$ converges by the Ratio Test, but to what value does it converge? To find out, complete the following steps.

Express $\frac{1}{1-x}$ as a series: $\frac{1}{1-x} =$

Differentiate both sides.

Multiply both sides by x.

Differentiate both sides (again).

Multiply both sides by x (again) and express the right-hand side in simplified Σ-notation

Substitute $\frac{1}{2}$ for x and simplify both sides of your equation. What result do you get?